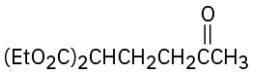
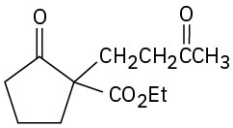
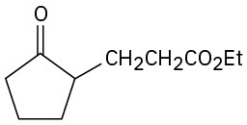
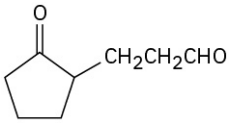
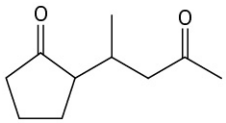
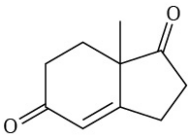
PROBLEM 23-17 **(a)**  **(b)** 

PROBLEM 23-18 $CH_3CH_2COCH{=}CH_2 + CH_3CH_2NO_2$

PROBLEM 23-19 **(a)**  **(b)**  **(c)** 

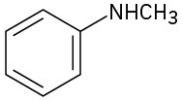
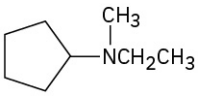
PROBLEM 23-20 **(a)** Cyclopentanone enamine + propenenitrile
(b) Cyclohexanone enamine + methyl propenoate

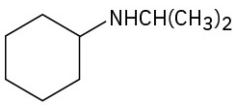
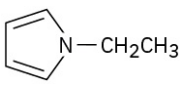
PROBLEM 23-21 

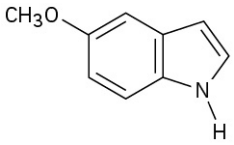
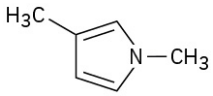
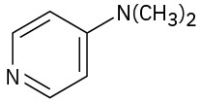
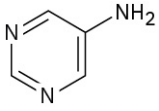
PROBLEM 23-22 2,5,5-Trimethyl-1,3-cyclohexanedione + 1-penten-3-one

Chapter 24

PROBLEM 24-1 **(a)** *N*-Methylethylamine **(b)** Tricyclohexylamine **(c)** *N*-Ethyl-*N*-methylcyclohexylamine **(d)** *N*-Methylpyrrolidine **(e)** Diisopropylamine **(f)** 1,3-Butanediamine

PROBLEM 24-2 **(a)** $[(CH_3)_2CH]_3N$ **(b)** $(H_2C{=}CHCH_2)_3N$ **(c)**  **(d)** 

(e)  **(f)** 

PROBLEM 24-3 **(a)**  **(b)**  **(c)**  **(d)** 

PROBLEM 24-4 **(a)** $CH_3CH_2NH_2$ **(b)** NaOH **(c)** CH_3NHCH_3

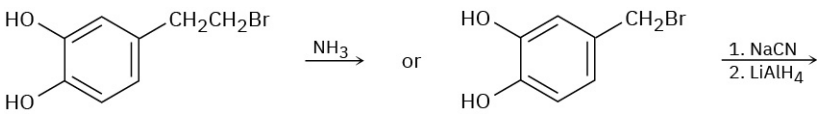
PROBLEM 24-5 Propylamine is stronger; benzylamine $pK_b = 4.67$; propylamine $pK_b = 3.29$

PROBLEM 24-6 **(a)** *p*-Nitroaniline < *p*-Aminobenzaldehyde < *p*-Bromoaniline
(b) *p*-Aminoacetophenone < *p*-Chloroaniline < *p*-Methylaniline
(c) *p*-(Trifluoromethyl)aniline < *p*-(Fluoromethyl)aniline < *p*-Methylaniline

PROBLEM 24-7 Pyrimidine is essentially 100% neutral (unprotonated).

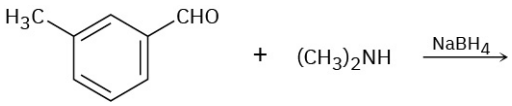
PROBLEM 24-8 **(a)** Propanenitrile or propanamide **(b)** *N*-Propylpropanamide **(c)** Benzonitrile or benzamide **(d)** *N*-Phenylacetamide

PROBLEM 24-9 The reaction takes place by two nucleophilic acyl substitution reactions.

PROBLEM 24-10 

PROBLEM 24-11 **(a)** Ethylamine + acetone, or isopropylamine + acetaldehyde **(b)** Aniline + acetaldehyde **(c)** Cyclopentylamine + formaldehyde, or methylamine + cyclopentanone

PROBLEM 24-12



PROBLEM 24-13 **(a)** 4,4-Dimethylpentanamide or 4,4-dimethylpentanoyl azide **(b)** *p*-Methylbenzamide or *p*-methylbenzoyl azide

PROBLEM 24-14 **(a)** 3-Octene and 4-octene **(b)** Cyclohexene **(c)** 3-Heptene **(d)** Ethylene and cyclohexene

PROBLEM 24-15 $H_2C{=}CHCH_2CH_2CH_2N(CH_3)_2$

PROBLEM 24-16 1. HNO_3, H_2SO_4; 2. H_2/PtO_2; 3. $(CH_3CO)_2O$; 4. $HOSO_2Cl$; 5. aminothiazole; 6. H_2O, NaOH

PROBLEM 24-17 **(a)** 1. HNO_3, H_2SO_4; 2. H_2/PtO_2; 3. 2 CH_3Br

(b) 1. HNO_3, H_2SO_4; 2. H_2/PtO_2; 3. $(CH_3CO)_2O$; 4. Cl_2; 5. H_2O, NaOH

(c) 1. HNO_3, H_2SO_4; 2. Cl_2, $FeCl_3$; 3. $SnCl_2$

(d) 1. HNO_3, H_2SO_4; 2. H_2/PtO_2; 3. $(CH_3CO)_2O$; 4. 2 CH_3Cl, $AlCl_3$; 5. H_2O, NaOH

PROBLEM 24-18 **(a)** 1. CH_3Cl, $AlCl_3$; 2. HNO_3, H_2SO_4; 3. $SnCl_2$; 4. $NaNO_2$, H_2SO_4; 5. CuBr; 6. $KMnO_4$, H_2O

(b) 1. HNO_3, H_2SO_4; 2. Br_2, $FeBr_3$; 3. $SnCl_2$, H_3O^+; 4. $NaNO_2$, H_2SO_4; 5. CuCN; 6. H_3O^+

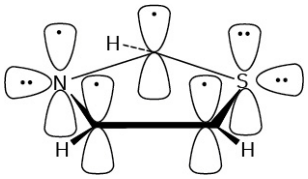
(c) 1. HNO_3, H_2SO_4; 2. Cl_2, $FeCl_3$; 3. $SnCl_2$; 4. $NaNO_2$, H_2SO_4; 5. CuBr

(d) 1. CH_3Cl, $AlCl_3$; 2. HNO_3, H_2SO_4; 3. $SnCl_2$; 4. $NaNO_2$, H_2SO_4; 5. CuCN; 6. H_3O^+

(e) 1. HNO_3, H_2SO_4; 2. H_2/PtO_2; 3. $(CH_3CO)_2O$; 4. 2 Br_2; 5. H_2O, NaOH; 6. $NaNO_2$, H_2SO_4; 7. CuBr

PROBLEM 24-19 1. HNO_3, H_2SO_4; 2. $SnCl_2$; 3a. 2 equiv. CH_3I; 3b. $NaNO_2$, H_2SO_4; 4. product of 3a + product of 3b

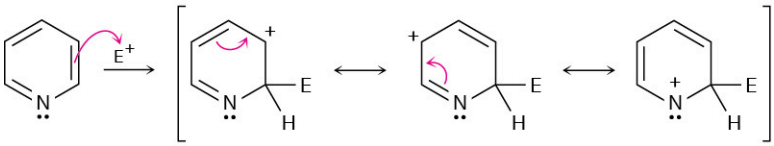
PROBLEM 24-20



PROBLEM 24-21 4.1% protonated

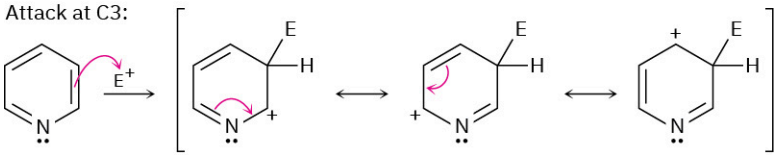
PROBLEM 24-22

Attack at C2:

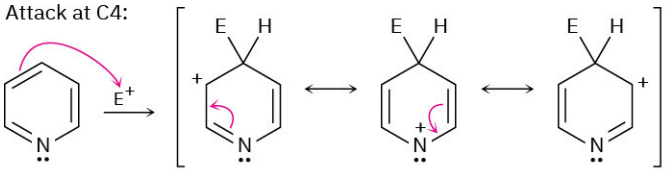


Unfavorable

Attack at C3:



Attack at C4:

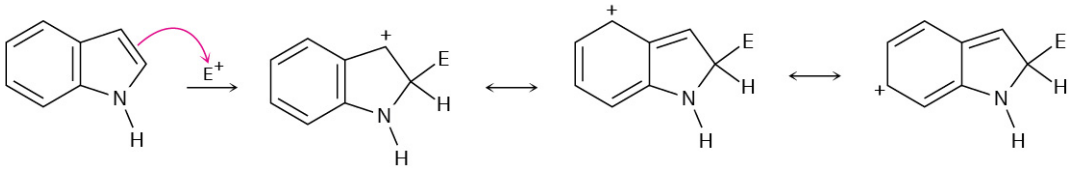


Unfavorable

PROBLEM The side-chain nitrogen is more basic than the ring nitrogen.

PROBLEM 24-23 Reaction at C2 is disfavored because the aromaticity of the benzene ring is lost.

PROBLEM 24-24



PROBLEM 24-25 $(CH_3)_3CCOCH_3 \rightarrow (CH_3)_3CCH(NH_2)CH_3$

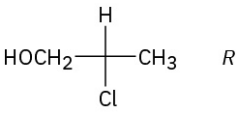
Chapter 25

PROBLEM 25-1 **(a)** Aldotetrose **(b)** Ketopentose **(c)** Ketohexose **(d)** Aldopentose

PROBLEM 25-2 **(a)** *S* **(b)** *R* **(c)** *S*

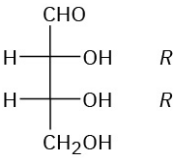
PROBLEM 25-3 A, B, and C are the same.

PROBLEM 25-4



R

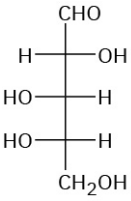
PROBLEM 25-5



R, *R*

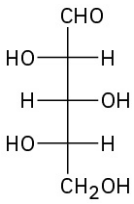
PROBLEM 25-6 **(a)** L-Erythrose; 2*S*,3*S* **(b)** D-Xylose; 2*R*,3*S*,4*R* **(c)** D-Xylulose; 3*S*,4*R*

PROBLEM 25-7

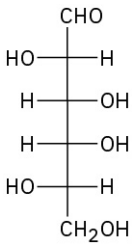


L-(+)-Arabinose

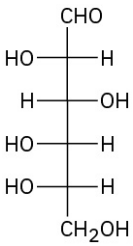
PROBLEM 25-8 **(a)**



(b)

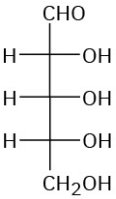


(c)



PROBLEM 25-9 16 D and 16 L aldoheptoses

PROBLEM 25-10



D-Ribose